# HARARE INSTITUTE OF TECHNOLOGY

## SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

### B.TECH (HONS) DEGREE IN INFORMATION SECURITY AND ASSURANCE

### PART 2 SEMESTER 2

### ISS 222: DATA COMMUNICATIONS AND NETWORKS

**TIME: 3 HOURS** **DATE: MAY/JUNE 2018**

**TOTAL MARKS: 100**

**INSTRUCTIONS TO CANDIDATES:**

- This question paper contains a total of **FIVE** questions.
- Each question carries **25** marks
- Answer **any FOUR** questions
- Start a new question on a fresh page

Question 1

a. With the aid of a fully labelled diagram, explain all the components involved in a data communication system. [5]

b. Explain the meaning of the following terms.

   a. Forward error correction (FEC) [3]

   b. Hamming distance [3]

   c. Piggybacking [3]

c. There are various categories of networks. Explain any three (3). [6]

d. Draw a fully labelled diagram of the IPv6 header. [5]

Question 2

a. Differentiate between TCP and UDP as they apply to layer four (4) of the OSI model. [5]

b. Describe in detail how DNS works with particular focus to recursive lookups. [10]

c. A co-worker calls the help desk complaining about her PC not going to the internet. Outline the procedure you would take in troubleshooting this problem. [10]

Question 3

You are attached to company ABC as an intern. The company is growing and there are four (4) departments namely Accounts, HR, IT and Sales & Marketing. To improve network performance you need to segment the network according to departments. The following table shows the number of employees associated with each department.

| Department | Number of Employees |
|---|---|
| Accounts | 10 |
| HR | 4 |
| IT | 4 |
| Sales & Marketing | 20 |

The company is using the following private IP address 192.168.0.0/24. You are required to

a. Design a subnetting scheme for the network that does not waste addresses and caters for an additional two members in each subnet. [10]

b. Define the following terms

   i. CSMA/CD [3]

   ii. CSMA/CA [3]

   iii. ARP [3]

c. Describe the TCP 3-way handshake using diagrams. [6]

Question 4

a. Describe the DHCP four step process with the aid of diagrams. [10]

b. Explain the working principle of stop and wait protocol [4]

c. Draw and explain the hierarchical structure of the Internet (ISPs). [6]

d. With the aid of an example describe the structure of MAC address. [5]

Question 5

a. With the aid of an example explain why TCP is called a reliable protocol. [6]

b. Differentiate between the OSI model and the TCP/IP protocol suite. [12]

c. Define the term VLAN giving its uses in your answer. [4]

d. Explain why a switch is said to be intelligent. [3]

********** END OF EXAMINATION **********